

THE UNIVERSITY OF SCIENCE
FACULTY OF INFORMATION TECHNOLOGY



Kiểm thử phần mềm - 22KTPM3
GUI & USABILITY TESTING - HW4

Thành Viên

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Chapter 1

Group Information

Group ID: 07

Member Name	Student ID	Screen	Status
Cao Uyển Nhi	22127310	Customer Registration Page	Done
Lưu Thanh Thuý	22127410	Checkout Page	Done
Nguyễn Phước Minh Trí	22127424	Categories Page	Done
Võ Lê Việt Tú	22127435	Home page	Done
Trần Thị Cát Tường	22127444	Categories Management Page	Done

Chapter 2

GUI Testing with AI tools

2.1 The Tool AI name and prompt used

In this report, we explore the use of Generative AI to enhance GUI (Graphical User Interface) testing by leveraging Grok, an AI-powered assistant developed by xAI. The objective of this activity is to assess the interface of our application against a predefined GUI checklist and identify any visual or functional inconsistencies. By inputting both the checklist and interface screenshots into Grok, we aim to evaluate its effectiveness in generating test cases and providing valuable feedback for UI compliance. This method allows us to compare AI-generated results with manually created test cases, validate their accuracy, and refine our GUI testing strategy based on Grok's suggestions.

- **AI Engine Used:** Grok 3 (built by xAI)
- **Platform:** [Website](#)
- **The prompt used:** Now I do the checklist creation step, here is the sample template and screenshot of the screen I will test. You are the test lead of a testing team Give me a GUI-testing checklist with the following columns: 'STT', 'Checkpoint', 'Yes', 'No', 'Remarks' for an application that:
 - Is a E-commerce website?
 - Focus on Tools product and hardware items
 - Use these main requirements for the checklist:
 - Consistency
 - Basic Application Functionality
 - Spelling Errors
 - Compatibility and Responsiveness
 - Typography
 - Forms, Fields, and Drop Downs
 - Validations
 - The Behavior of UI Elements

- Spacing
- Images

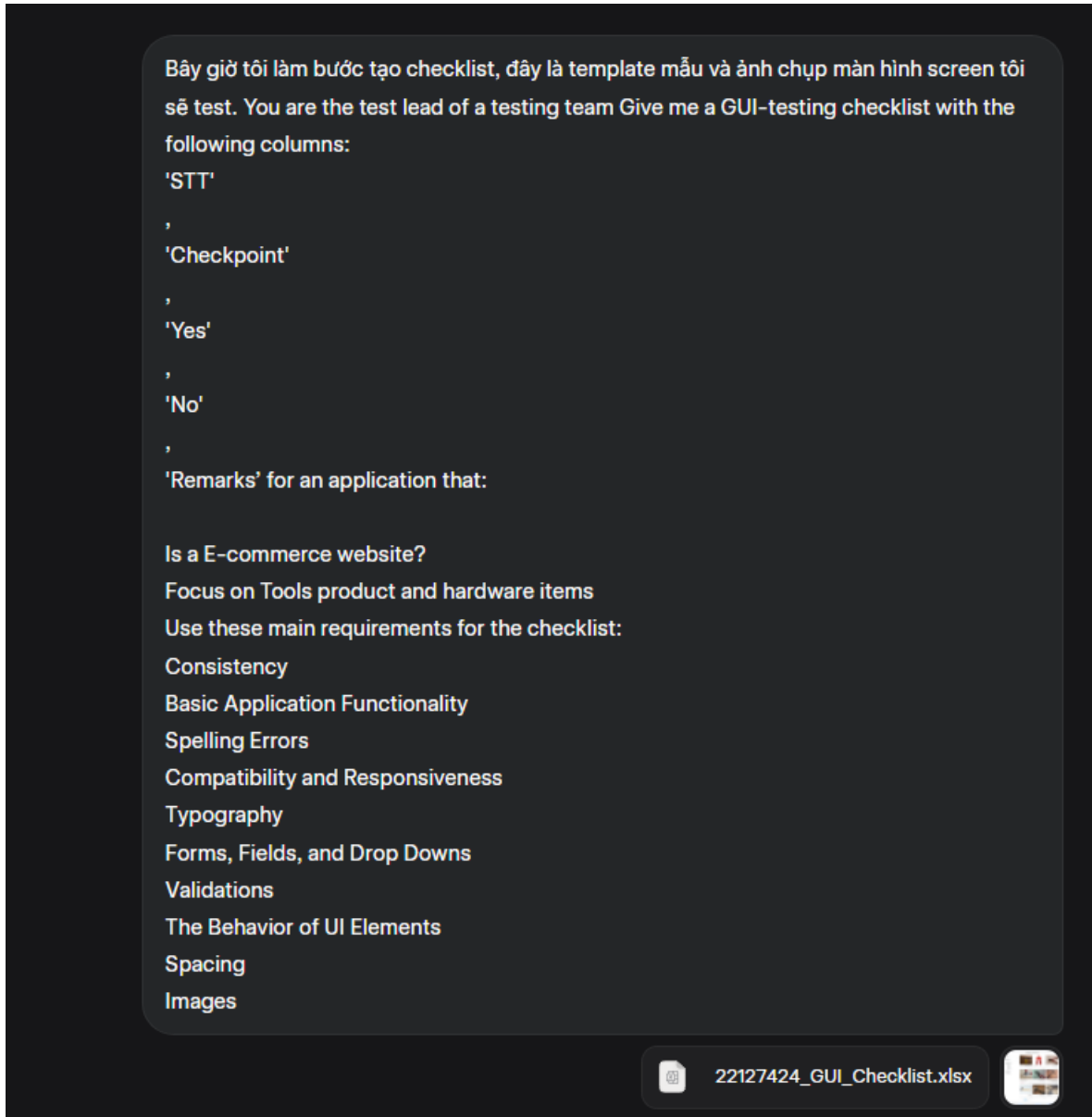


Figure 2.1: Prompt used

2.2 Validated and refined the AI-generated results

To ensure the accuracy of the issues identified by AI (e.g., Grok 3), follow these steps to validate and refine the results:

- **Step 1: Manual Verification**

- Access the The Toolshop application (version /sprint5-with-bugs) directly and navigate to the “Category: Hand Tools” page.

- Perform manual checks on each issue reported by AI based on the screenshot and analysis results. For example: “Manually verified the issues reported by AI on the ‘Category: Hand Tools’ page by directly accessing the The Toolshop application
- **Step 2: Confirm Specific Issues**
 - Use manual observation or tools to validate each reported issue.
- **Step 3: Refine or Discard Suggestions**
 - If AI results are inaccurate, refine or discard them based on manual findings. For example: “Discarded the suggestion ‘Previous button does not work’ as manual testing on Chrome

2.3 Which test cases came from AI and which were created manually

Test case	Source
Section 1 (1.1, 1.2, 1.3, 1.4,)	Self-created
Section 1 (1.5)	AI
Section 2 (2.1)	Self-created
Section 2 (2.2, 2.3)	AI
Section 3	Self-created

2.4 Conclusion

Integrating Grok 3 into GUI testing workflows offers a practical and efficient approach to detecting visual and structural UI issues early in the development process. By blending AI-driven analysis with checklist validation, I achieved a balance of speed and precision in identifying critical interface defects. This experiment highlights that AI serves as a valuable co-pilot in GUI assessment, particularly when supported by structured human guidance.

Chapter 3

Self-Evaluation

Criteria	Max Point	Self-assessment
Checklist of one GUI	30	30
AI tools	20	20
Bug report	30	30